CES 2025: A Silicon-Sampling Test

Ten questions · fifteen groups · CES 2021 profiles · two hosted models

2026-10-03

1. What we are testing

Can an LLM, given a profile of a Canadian, reproduce how Canadians answered ten CES 2025 questions? Does it help to give it a richer profile or a bigger model?

Strict temporal holdout. CES 2025 is used **only as the answer key** for scoring. Profiles and all baselines come from **CES 2021**. Nothing about 2025 respondents, their answers or their answer distributions is shown to any model or used to build any comparator.

Scoring is at the group level only. *Five regions × three ages = 15 groups.* In each group, the model's 100 answers are compared with the 2025 human answer distribution. Synthetic rows are built from 2021 respondents and have no 2025 counterpart, so there is no per-person accuracy.

Regions: BC; AB+SK+MB+Terr (Alberta, Saskatchewan, Manitoba, Yukon, Northwest Territories, Nunavut); ON; QC; ATL (New Brunswick, Nova Scotia, Prince Edward Island, Newfoundland and Labrador). **Ages:** 18–34, 35–54, 55+. Groups are assigned from CES age and province, identically in 2021 and 2025.

2. Ten questions, identical across all runs

English campaign-period items; prefix variable suffixes with `cps25_`. Full wording and options: `questions.json`.

1. How much should the federal government spend on **affordable housing**? (`spend_afford_h`)
2. How much should the federal government spend on **a national childcare system**? (`spend_nation_c`)
3. How much should the federal government spend on **defence**? (`spend_defence`)
4. How much should the federal government spend on **reconciliation with Indigenous Peoples**? (`spend_rec_indi`)
5. Individuals who are terminally ill should be allowed to end their lives with the assistance of a doctor. (`pos_life`)
6. The federal government should do more to help build oil pipelines. (`pos_energy`)
7. When there is a conflict between protecting the environment and creating jobs, jobs should come first. (`pos_jobs`)
8. Do you think Canada should admit: **more, fewer, or about the same number of immigrants**? (`imm`)
9. On the whole, how satisfied are you with the way democracy works in Canada? (`demsat`)
10. Over the past year, has your financial situation: **got better, stayed about the same, or got worse**? (`own_fin_retro`)

Items 1–4 offer less / same / more spending; 5–7 offer five agreement levels; 9 offers four satisfaction levels. **Every item also offers "Don't know/Prefer not to answer"; retain it as an answer category.** Do not collapse options.

2021 availability (verified 2026-10-03). All ten items exist in CES 2021 (`cps21_*`), with the same answer options; only the punctuation of "Don't know/ Prefer not to answer" differs, which is normalized. Nine items have identical wording. **One differs: `pos_energy`.** 2021 asks whether the government "should do more to help Canada's energy sector, including building oil pipelines"; 2025 asks only about building oil pipelines. It is used for training and baselines as a near-match and reported separately. Module split in 2021: housing and childcare spending each went to about half the respondents (n 10.5k each), the three agreement items to about 12.5k, and the other five to everyone (n = 20,968).

3. Design: 6 inference runs

	Basic	Rich	Rich + partisan
Small model (Qwen3.5-9B), Together	run 1	run 2	run 3
Large model (Qwen3.8-2.4T-A95B), Together	run 4	run 5	run 6

Each run is **1,500 calls, one per profile, each answering all ten questions** (100 profiles × 15 groups).

- **Small model: `Qwen/Qwen3.5-9B`.** It is the only 7–9B model Together serves pay-per-call (checked 2026-10-03: 16 of 74 priced chat models answered without a dedicated endpoint). Price: $0.17 / $0.25 per M input/output tokens.
- **Large model: `Qwen/Qwen3.8-2.4T-A95B`** (mixture of experts: 2.4T total, 95B active parameters). It is from the same family as the small model, so the size comparison is not confounded by model family. Price: $2 / $6 per M tokens.
- **Thinking off.** Both models think by default and return an empty answer within a short token limit. Pass `chat_template_kwargs: {enable_thinking: false}` on Together. The 2.4T model sometimes adds a disclaimer ("I don't have personal opinions…"); the parser reads the ten `n: k` lines and ignores other text.
- **Cost estimate** (measured in the smoke test: about 900 input and 50–65 output tokens per call, 4,500 calls per model): about $0.75 for the 9B and $9 for the 2.4T; about $10 in total.
- **Contamination.** Both models were released in 2026, after CES 2025 fieldwork. They may know the 2025 campaign and even published CES 2025 results. The only pay-per-call model with a pre-2025 cutoff is `Llama-3.3-70B-Instruct-Turbo`, and it has no small sibling on Together. Report this as a limitation rather than switching.
- **Fine-tuning deferred.** A local LoRA fine-tune of the 9B on CES 2021 (plus a matching local 4-bit base, to separate fine-tuning from quantization) is out of scope for now: a local smoke test crashed the 16 GB M1. It can be added later as runs 7–12 without changing anything below, since profiles, prompt and scoring are shared.

4. Profiles: drawn from CES 2021

Draw once, reuse everywhere. For each of the 15 groups, draw **100 CES 2021 campaign-period respondents** without replacement, using a fixed seed. Save them in `profiles_2021.csv`: synthetic run ID, CES 2021 respondent ID, group, all profile fields. All runs and all baselines use these **same 1,500 people**, so conditions differ only in which fields are shown.

Conditions (fields cumulative):

- **Basic:** region (as the plain-language description above), age band, gender.
- **Rich:** basic + province, education, household income, first language, religion, born in Canada.
- **Rich + partisan:** rich + federal party identification, left–right self-placement (0–10), 2019 vote recall. **Never** include answers to any of the ten test questions or close substitutes.

CES 2021 sources, all from the campaign-period survey: `cps21_age`, `cps21_genderid`, `cps21_province`, `cps21_education`, `cps21_income_cat`, `cps21_language_*`, `cps21_religion`, `cps21_bornin_canada`, `cps21_fed_id`, `cps21_lr_scale_bef_1`, `cps21_vote_2019`. Urban/rural is left out: it was asked only after the election (`pes21_rural_urban`) and is missing for 28% of respondents. Exact recodes are fixed in `build.py`. A missing profile field is shown as **"Not stated"**, so respondents are not dropped. Profiles describe the 2021 sample's composition within each group; if it differs from 2025's, that is a stated limitation.

5. Prompt and inference settings

One fixed template for every model and run, stored in the repo (`prompt.txt`) and filled from `profiles_2021.csv` and `questions.json`:

- system message: a short neutral instruction (simulated survey respondent; answer every question; option numbers only);
- user message: the profile as a bulleted list, then the ten questions **in the fixed order above**, each with its numbered options;
- required output: ten lines `1: <option number>` … `10: <option number>`.

Settings: **temperature 1.0, top-p 1.0**, fresh context per call, no examples, no 2025 information. Reasoning/"thinking" modes off where the model offers them; `max_tokens` 1024, so a reply that reasons anyway can still finish (the parser reads only text after a closing reasoning tag). A missing, unparseable or out-of-range answer becomes **Invalid output** for that item only. Retry only failed HTTP requests, never undesired answers. Record model ID, provider, date, settings and token counts in `runs.json`.

Output: `results_<model>_<condition>.csv`, the same 13-column schema as `validation.csv` (synthetic run ID, ten answers as exact option text, Age, Region), 100 rows per group.

6. Validation file and exact TV calculation

`validation.csv`: 20,180 CES 2025 respondents, 13 columns: `cps25_ResponseId`, the ten question variables, `Age`, `Region`. Answers are readable option labels. `Not asked / missing` is excluded per question; don't-know answers are kept.

Let j be a question, g a group and k an answer category. n_{jg} is the number of 2025 humans in g who answered j. Using raw counts:

$$p_{jgk} = \frac{\text{human count choosing } k \text{ in } (j,g)}{n_{jg}}, \qquad q_{jgk} = \frac{\text{model count choosing } k \text{ in } (j,g)}{100}, \qquad \boxed{TV_{jg} = \frac{1}{2}\sum_k |p_{jgk} - q_{jgk}|}$$

Invalid output is an extra category with human probability zero, and the denominator stays 100. That gives **150 cell TVs per run**. Flag cells with $n_{jg} < 30$. Stop if a cell has no human answers.

Overall TV per question. With question-specific group shares $a_{jg} = n_{jg}/N_j$ (`group_weights.csv`), aggregate first, then compute TV:

$$P_{jk} = \sum_g a_{jg} p_{jgk}, \qquad Q_{jk} = \sum_g a_{jg} q_{jgk}, \qquad \boxed{TV_j^{\text{overall}} = \frac{1}{2}\sum_k |P_{jk} - Q_{jk}|}$$

The target is the observed CES 2025 sample, not a survey-weighted population estimate. Overall TV is not the mean of cell TVs. **Headline score per run: the mean of the ten overall TVs.** Uncertainty: 95% intervals from 1,000 bootstrap resamples of the 100 synthetic rows within each group.

7. Baselines (all from CES 2021)

Each baseline produces one answer per profile in `profiles_2021.csv` (1,500 rows) and is scored exactly like a model.

1. **Overall mode:** the most common 2021 answer to each question, given to everyone (ties: smallest CES code).
2. **Group mode:** the most common 2021 answer within each region × age group.
3. **Multinomial logit, one per condition:** fitted on all other CES 2021 respondents (excluding the 1,500 profile respondents), using exactly that condition's profile fields, one model per question. For each profile, draw one answer from its predicted probabilities (fixed seed). This shows whether an LLM beats a simple statistical model given the **same information**.

Baselines 1–2 ignore the profile details. Baseline 3 is the main comparator for each condition. All baselines carry 2021→2025 opinion change as error. That drift is part of what is being tested.

8. What we report

- Headline score with CI for all 6 runs and all baselines; ten overall TVs and 150 cell TVs per run.
- **Improvement = baseline TV − model TV** (positive means the model wins), against overall mode and the matching logit, per question and per cell. Report how many of the 150 cells improve.
- Effects: **richness** (basic → rich → rich + partisan, within model), **size** (runs 1–3 vs. 4–6).
- Invalid-output rate per run and item; `pos_energy` (reworded since 2021) reported separately.
- Contamination note: each model's release date and training cutoff relative to CES 2025 fieldwork. Synthetic results are not polling results.

9. Order of work

1. CES 2021 v2.0 is in `opinionbench-research/data/raw/CES2021/` (n = 20,968). Record its SHA-256 (`d8abbff6…`) in `manifest.json`.
2. Ten items and profile variables verified (§2, §4); write the recodes.
3. Models picked (§3). Fine-tuning deferred.
4. Built `profiles_2021.csv`, `prompt.txt` and baselines; dry-run with the mock backend; 15-call live smoke test per model.
5. Runs 1–6 on Together (2026-10-03); scored once with `score_all.py`.

10. Results (runs of 2026-10-03)

Headline: mean of the ten overall TVs (lower is better; 95% bootstrap interval in brackets). Total cost about $11 ($0.70 for the 9B, $10.40 for the 2.4T).

	Basic	Rich	Rich + partisan
Qwen3.5-9B (runs 1–3)	0.357 [0.351, 0.364]	0.323 [0.317, 0.330]	**0.258** [0.251, 0.266]
Qwen3.8-2.4T-A95B (runs 4–6)	0.477 [0.471, 0.483]	0.421 [0.416, 0.427]	0.323 [0.316, 0.330]
2021 multinomial logit, same fields	0.161 [0.155, 0.171]	0.155 [0.149, 0.165]	0.150 [0.144, 0.160]
Cells (of 150) where the LLM beats the matching logit: 9B / 2.4T	9 / 6	15 / 7	26 / 14

2021 overall mode: 0.615; 2021 group mode: 0.504. Every LLM run beats both. Invalid output: 0.2–1.2% of answers per run; 29 of the 4,500 large-model calls hit the 1,024-token limit after reasoning despite thinking being off.

Overall TV by question

Question	9B basic	9B rich	9B partisan	2.4T basic	2.4T rich	2.4T partisan	Logit partisan	2021 mode
`spend_afford`	0.21	0.18	0.06	0.25	0.26	0.11	0.05	0.28
`spend_nation`	0.39	0.43	0.27	0.46	0.50	0.37	0.08	0.60
`spend_defence`	0.43	0.45	0.24	0.44	0.47	0.31	0.36	0.71
`spend_rec_indiv`	0.68	0.61	0.42	0.70	0.68	0.52	0.16	0.78
`pos_life`	0.19	0.14	0.11	0.40	0.20	0.19	0.05	0.52
`pos_energy` (reworded since 2021)	0.43	0.33	0.27	0.52	0.56	0.46	0.16	0.73
`pos_jobs`	0.14	0.14	0.25	0.55	0.52	0.39	0.13	0.79
`imm`	0.62	0.59	0.57	0.55	0.58	0.43	0.33	0.74
`demsat`	0.24	0.21	0.19	0.43	0.26	0.26	0.05	0.48
`own_fin_retro`	0.24	0.15	0.19	0.49	0.17	0.18	0.13	0.52

Findings

- **Neither LLM approaches the 2021 logit given the same information.** The best run (9B, rich + partisan) has about 1.7 times the logit's TV and beats it in 26 of 150 cells. The only question where an LLM beats the logit overall is defence (9B partisan 0.24, 2.4T partisan 0.31, logit 0.36).
- **Richness helps** for both models, with non-overlapping intervals at each step. Partisan fields give the largest gain. It is not uniform by question: the 9B gets worse on `pos_jobs` when partisan fields are added.
- **Size hurts.** The 2.4T model is worse than the 9B in every condition. Its answers are more concentrated than the human distribution: given basic profiles, 78% say "about the same number of immigrants" and 75% "spend about the same on defence".
- **Systematic misses, not just low variance.** On immigration, 62% of 2025 respondents chose "fewer"; the 9B chose it 2–4% of the time and the 2.4T 12–23%. On reconciliation spending, the models chose "spend more" 68–91% of the time against 22% of humans; on defence they favour "about the same" while 56% of humans chose "spend more". Defence and immigration are the items that moved most from 2021 to 2025 (TV 0.35 and 0.33 between the 2021 and 2025 distributions; all others 0.04–0.17), which is also why the logit does worst on them. Reconciliation moved less (0.17), so the models' miss there is not explained by drift.
- **Contamination.** Both models were released after CES 2025 fieldwork. The immigration result gives no sign that they reproduce 2025 opinion, but this is not a test of contamination.